Global Facilities – Design & Construction Standard – I&C – Fire Alarm and HSSD System

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1.0 Purpose

This Design & Construction Standard defines the minimum requirements for design, supply, packing and delivery of Fire Alarm System and High Sensitivity Smoke Detection System to greenfield project, including termination of fire alarm cables, factory & site testing, and commissioning.

Any deviation from this standard shall be submitted for approval during evaluation. Deviation list format shall refer to Appendix A in this standard document.

2.0 Scope

2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects

2.2 Following are in the scope of Fire Alarm system vendor:

1. Preparation of the layout drawings.
2. The vendor shall provide cable schedule / MTO complete with all details such as Type, Size, Length & Quantity of all cables.
3. Provision of clean room / hazardous area type devices (as applicable).
4. Liaison with other system vendors for necessary interfacing.
5. Program software.
6. Operation & Maintenance spares.
7. As Built Drawings.
8. Authority submissions.
9. Operation and Maintenance Manual.
10. Functional Matrix and Sequence of Operation.

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11. Factory Acceptance Test.
12. Loop check
13. Site Acceptance Test.
14. Training

3.0 Definitions and Acronyms

- **AC** - Alternating Current
- **DC** – Direct Current
- **AWG** - American Wire Gauge
- **DC** – Direct Current
- **FACU** – Fire Alarm Control Unit
- **FAS** – Fire Alarm System
- **HVAC** – Heating, Ventilating, and Air Conditioning
- **IR** – Infrared
- **LED** – Light Emitting Diode
- **NEC** – National Electric Code
- **NFPA** – National Fire Protection Association
- **OS** – Operating System
- **PC** – Personal Computer
- **SPDT** – Single-pole, double-throw
- **UL** – Underwriters Laboratories Inc.
- **UV** – Ultraviolet
- **UV/IR** – Ultraviolet / Infrared
- **HSSD** – High Sensitivity Smoke Detection
- **FM** – Factory Global (Factory Mutual)
- **HLI** – High Level Integration
- **FACP** – Fire Alarm Control Panel
- **BSGS** – Bulk Specialty Gas System
- **LCD** – Liquid Crystal Display
- **LED** – Light Emitting Diode
- **RFI** – Radio Frequency Interference
- **EMC** – Electromagnetic Compatibility

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4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learnt and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 General Description

5.1 Applicable Codes and Standards

The fire alarm system and the components used shall conform to the latest edition of the following codes and standards:

[Singapore]

1. SS645, 2019, Code of Practice for Installation and Servicing of Electrical Fire Alarm Systems.
2. FSB Code, Code of Practice for Fire Precautions in Buildings, 2007, Fire Safety Bureau, Singapore Civil Defense Force.
3. SS299, Singapore Standard 299 for Fire Resistant Cables: Part-1.
4. CP5, 2009, Code of Practice for Electrical Installation with amendment.
5. Fire Engineering Study

[Taiwan]

6. Standard for Installation of Fire Safety Equipment Based on Use and Occupancy, dated 2018.10.17

[Japan]

7. Fire Service Act, Act No. 186 of July 24, 1948

[All]

8. International Building Code (IBC)
9. International Fire Code (IFC) 2012
10. International Mechanical Code (IMC) 2012
11. NFPA 70 – National Electric Code
12. NFPA 72 – National Fire Alarm Code 2013
13. NFPA 75 – Protection of Computer / Data Processing Equipment

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14. NFPA 90A – Air-Conditioning Systems
15. NFPA 92A – 2000 Smoke Control Systems
16. NFPA 92B – 2000 Smoke Management Systems in Malls, Atria, and Large Areas.
17. NFPA 318 – Protection of Clean room
18. NFPA 101 – 2000 Fire Alarm Code®.
19. NFPA 12 - 2000 Carbon Dioxide Extinguishing Systems
20. NFPA 13 - Installation of Sprinkler Systems
21. NFPA 15 - 1996 Water Spray Fixed Systems for Fire Protection
22. NFPA 17 - 1998 Dry Chemical Extinguishing Systems
23. NFPA 2001 - 2000 Clean Agent Fire Extinguishing Systems
24. UL 864 - Control Units for Fire Protective Signaling Systems
25. UL 268 - Smoke Detectors for Fire Protective Signaling Systems
26. UL 268A - Smoke Detectors for Duct Applications
27. UL 217 - Single and Multiple Station Smoke Alarms
28. UL 521 - Heat Detectors for Fire Protective Signaling Systems
29. UL 228 - Door Closers-Holders, With or Without Integral Smoke Detectors
30. UL 464 - Audible Signaling Appliances
31. UL 38 - Manually Actuated Signaling Boxes for Use with Fire-Protective Signaling Systems
32. UL 346 – Water-flow Indicators for Fire Protective Signaling Systems
33. UL 1971 - Signaling Devices for the Hearing-Impaired
34. UL 1481 - Power Supplies for Fire Protective Signaling Systems
35. UL 1711 - Amplifiers for Fire Protective Signaling Systems
36. UL 1635 - Digital Alarm Communicator System Units
37. Americans with Disabilities Act (ADA)
38. American National Standards Institute 117.1 (ANSI)
39. Factory Mutual (FM) approval
40. IEC-60331, Tests for Electrical Cables under Fire Conditions.
41. International Standards Organization (ISO) 35). ISO-9000
42. ISO-9001

5.2 System Description

System to be a fully functional fire alarm system and emergency communication system acceptable to the local authority having jurisdiction and in accordance with applicable codes and standards consisting of the following elements:

1. The fire alarm system supplied is an addressable, intelligent, peer-to-peer, node to node networked fire alarm control units. The system utilizes independently addressable initiating devices, input/output control modules, visual notification appliances, audible notification appliances, and auxiliary devices.
2. The emergency communication system provides the command and control to indicate the existence of an emergency and broadcasting

information necessary to facilitate an appropriate response and action.

3. A fiber optic communication network allows data and audio communication exchange between fire alarm control units and fire alarm operator workstations.
4. Personal Computer (PC)-based operator station(s) and software programming, functioning as the primary operator interface for the fire alarm and emergency system.
5. A fire command center provides the monitoring of the status, disposition of signals, and control for the fire alarm system and emergency communication system.
6. Integral emergency voice/alarm communications with voice intelligibility.

6.0 Design

6.1 Design Criteria.

The following criteria to be followed:

1. The fire alarm system and emergency communication system are designed and installed to support zero network downtime.
2. A graphical command workstation is integral part of the fire command centers and serves as the operational and administration functions required for the fire alarm system. The workstation shall be located one in facility control room and one in Fire Command Center room.
3. Each addressable loop to have remaining capacity of 20 percent after site acceptance testing.
4. Fiber optic networks shall comply with NFPA 70, Section 770.
5. Addressable initiating device circuits to be 16 American Wire Gauge (AWG) or 1.5mm, unless otherwise subject to approval.
6. Where 24 Vdc power is being supplied from the fire alarm panels, panel design to allow de-energization of the panel (for maintenance) without affecting door holder and smoke damper control circuits.
7. Fire alarm circuits to be installed in dedicated fire alarm conduit or tray only throughout the facility.
8. Activation of any initiating device shall print out a 32-character message at the fire alarm control panel and fire alarm workstation. History is recorded electronically on the Fire alarm computer.
9. Activation of any duct smoke detector associated with a particular air handling unit will shut down the associated fan/s in that unit. Waiver shall be raised to the authority if the activation has production impact, subject to EHS review.
10. Each duct smoke detector interface requirement shall comply with local code.
11. Activation of any manual pull station or a sprinkler flow switch shall activate a fire alarm immediately. The fire alarm shall go continuous until the alarm is acknowledged and reset.
12. Upon receipt of an alarm activation signal from the Fire alarm control panel, all sub alarm panels shall activate the alarm bells and strobes.

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13. Each sprinkler system tamper switch and each pre-action system tamper or low air supervisory switch shall report to the Fire Alarm system as a supervisory.
14. Supervisory Operation: Activation of a supervisory condition shall cause the internal audible device to sound at the control panel and command center, display the event on the graphical workstation and display a pictorial image. The LCD display shall indicate all applicable information associated with the supervisory condition including the following: zone, device type, device location, device address, time, and date. All system activity/events shall be documented in the system. Any remote or local annunciator associated with the supervisory zone shall be illuminated. Transmit signal to the central station with point identification.
15. Trouble Operation

    Activation of a trouble condition shall cause the internal audible device to sound at the control panel and command center. Display the event on the graphical workstation and display a pictorial image. The LCD display shall indicate all applicable information associated with the trouble condition including zone, device type, device location, device address, time and date. Any remote or local annunciator associated with the trouble zone shall be illuminated. Transmit signal to the graphic workstation with point identification.

6.2 Submittal Requirement

1. Design Calculations which will include:
   i. The system shall be designed for interior building audibility which shall comply with NFPA 72 or local code.
   ii. Power Supply Battery Calculations: Provide battery capacity calculations for each power supply that uses batteries for secondary power. Identify all loads. Identify any loads shed during alarm operation. Use the manufacturer’s recommended methods and/or forms.
   iii. 24 VDC Notification Appliance Circuits: Provide voltage drop calculations using the manufacturer’s recommended methods and/or forms.
2. Manufacturer’s catalog data, to include material description, agency approval, and listing.
3. The system vendor shall furnish Sequence of Operation including functional matrix or cause effect matrix.
4. Shop Drawings shall meet the following requirements:
   i. System schematic diagrams which individually depict all control panels, annunciators, addressable devices, and notification appliances. Diagrams include specific addresses that corresponds to addresses shown on the floor plans. Drawing shall provide wire specifications, and wire identification for all conductors depicted on the schematic diagram. All circuits shall have identifiers that correspond with those required on the control panel and floor plan drawings. End-of-line resistors and values shall be depicted.
   ii. Provide device floor plans for all areas served by the fire alarm system. Floor plans shall indicate accurate locations for all control

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and peripheral devices. All addressable devices are shown. Coordinate the devices address with the same device shown on the schematic diagram.

iii. Identify all notification appliance with a circuit number and item number. Coordinate the circuit and item number with the same device shown on the system schematic diagram.
iv. Control panel drawings shall show internal component placement and all internal and field terminations. Provide details indicating where conduit connections shall be made to avoid conflicts with internally mounted batteries. For each additional fire alarm panel, a separate drawing which clearly indicated the panel designation, service and location of the control enclosure.
v. When addressable module is used in multiple configuration for monitoring or controlling equipment, provide a drawing for each application.
vi. Provide a fire alarm system functional matrix that illustrates alarm input/output events in association with the initiation devices. Matrix summary shall include system supervisory and trouble output functions.
vii. Provide matrix that illustrate the initiation input address and the control module output address for the emergency control functions for HVAC systems, elevator recall and door releasing service.
viii. Provide startup and commissioning forms.

5. Other relevant documents or drawings which are required by local authority.

7.0 Quality Assurance

1. Material and equipment shall be new and shall confirm to grade, quality and standards specified. Equipment or materials of same type shall be product of same manufacturer throughout. The manufacturer of the Fire Alarm system shall have minimum of 10 years documented experience in the design and construction of similar equipment.
2. Materials and equipment shall confirm to international standards as listed in codes and standards.
3. Provide the work in accordance with local code.

   Where required by Authority, material and equipment shall be listed and labelled by National recognized testing laboratory or other organization acceptable by the authority.
4. Provide materials and equipment acceptable to local Authority for Class, Division, and Group of hazardous area indicated.
5. Material and equipment must be listed and labelled for its intended purpose, environment, or application, especially when used in extreme climate areas.

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8.0 Fire Alarm Component

8.1 Addressable Heat Detector

1. Heat detector shall be of electronic type for combined rate-of-rise and fixed temperature response. It may consist of two independent thermistors, designed to automatically compensate changes in ambient conditions.
2. All electronic circuits must be solid state, using Surface Mounted Device Technology (SMD) and the entire circuitry must be hermetically sealed to prevent the operation being impaired by dirt, dust or humidity.
3. Addressable devices shall provide an address-setting means using rotary decimal switches. Because of the possibility of installation error, systems that use binary jumpers or DIP switches to set the detector address are not acceptable. Addressable devices that require the address be programmed using programming utility are not acceptable. The detectors shall also store an internal identifying code that the control panel shall use to identify the type of detector.
4. The detector shall be protected against damage by reverse of polarity or faulty wiring. The detector shall have a built-in line isolator which shall be able to automatically isolate short circuits on the detector line such that any single short circuit in the detection loop will not impair the operation of the system or detectors.
5. Addressable heat detectors shall provide dual alarm and power/polling LEDs. Both LEDs shall flash green under normal conditions, indicating that the detector is operational and in regular communication with the control panel, and both LEDs shall be placed into steady red illumination by the control panel, indicating that an alarm condition has been detected. If required, the LED flash shall have the ability to be removed from the system program. An output connection shall also be provided in the base to connect an external remote alarm LED.
6. The fire alarm control panel shall permit detector sensitivity adjustment through field programming of the system. The panel on a time-of-day basis shall automatically adjust sensitivity.
7. Using software in the Fire Alarm Control Panel, detectors shall automatically compensate for dust accumulation and other slow environmental changes that may affect their performance. The detectors shall be listed by UL as meeting the calibrated sensitivity test requirements of NFPA Standard 72 or equivalent.
8. The heat detector must be immune to external Electro-Magnetic Interference (EMI) generated by radio transmission equipment or power lines. It shall have EMI protection up to 50 Volts/meter electric field strength tested in accordance to IEC61000-4-3.
9. The system design shall allow a single pair of wires to access multiple loops containing addressable devices. All heat detectors shall be suitable for direct connecting to the class "A" four wires individual addressable loop. The detector shall be inserted into or removed from the base by a simple push-twist mechanism. It shall be possible to remove / exchange detector up to 7 meter above floor level by operator with special tool.

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10. The detector must have a built-in response indicator and shall have the possibility of connecting another freely programmable remote indicator to indicate the alarm status of other detector or zone.
11. Intrinsically safe addressable heat detectors shall be used for detectors installed within hazardous areas.
12. The detector shall be individually identifiable, with all relevant data, throughout its lifetime by a specific serial number and other corresponding information, readable at the control unit with authorization password. The data shall be stored in a non-volatile memory inside the detector. All heat detectors shall be common and interchangeable to a common type of base.
13. The heat detector shall be IP44 rated as a minimum.
14. Systems without built in isolator shall provide Isolators separately for each device.

8.2 Addressable Smoke Detector

1. The smoke detector shall be of high-performance type which uses an optical smoke sensor or a combined optical smoke sensor and a heat sensor. Its evaluation criteria shall be based on the smoke density, variation of smoke density over time, rate-of-rise of temperature and maximum temperature.
2. All electronic circuits must be solid state, using Surface Mounted Device Technology (SMD) and the entire circuitry must be hermetically sealed to prevent the operation being impaired by dirt, dust or humidity.
3. Addressable devices shall provide an address-setting means using rotary decimal switches. Because of the possibility of installation error, systems that use binary jumpers or DIP switches to set the detector address are not acceptable. Addressable devices that require the address be programmed using programming utility are not acceptable. The detectors shall also store an internal identifying code that the control panel shall use to identify the type of detector.
4. The detector shall be protected against damage by reverse of polarity or faulty wiring. The detector shall have a built-in line isolator which shall be able to automatically isolate short circuits on the detector line such that any single short circuit in the detection loop will not impair the operation of the system or detectors.
5. Addressable smoke detectors shall provide dual alarm and power/polling LEDs. Both LEDs shall flash green under normal conditions, indicating that the detector is operational and in regular communication with the control panel, and both LEDs shall be placed into steady red illumination by the control panel, indicating that an alarm condition has been detected. If required, the LED flash shall have the ability to be removed from the system program. An output connection shall also be provided in the base to connect an external remote alarm LED.
6. The fire alarm control panel shall permit detector sensitivity adjustment through field programming of the system. The panel on a time-of-day basis shall automatically adjust sensitivity.

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7. Using software in the Fire Alarm Control Panel, detectors shall automatically compensate for dust accumulation and other slow environmental changes that may affect their performance. The detectors shall be listed by UL as meeting the calibrated sensitivity test requirements of NFPA Standard 72 or equivalent.
8. The smoke detector shall be designed to suppress transient interference and other deceptive phenomena's without impairing the capability of detecting real smoke and shall be conform to NFPA 72 or equivalent.
9. The smoke detector must be immune to external Electro-Magnetic Interference (EMI) generated by radio equipment. It shall have EMI protection up to 50 Volts/meter electric field strength tested in accordance to IEC 61000-4-3.
10. The system design shall allow a single pair of wires to access multiple loops containing addressable devices. All smoke detector shall be suitable for direct connecting to the class “A” four wires individual addressable loop. The detector shall be inserted into or removed from the base by a simple push-twist mechanism. It shall be possible to remove / exchange detector up to 7 meter above floor level by operator with special tool.
11. The detector must have a built-in response indicator and shall have the possibility of connecting another freely programmable remote indicator to indicate the alarm status of other detector or zone.
12. Intrinsically safe addressable heat detectors shall be used for detectors installed within hazardous areas.
13. The detector shall be individually identifiable, with all relevant data, throughout its lifetime by a specific serial number and other corresponding information, readable at the control unit with authorization password. The data shall be stored in a non-volatile memory inside the detector. All detectors shall be mounted on a common and interchangeable base.
14. The smoke detector shall be IP44 rated as a minimum.
15. Systems without built in isolator shall provide isolators separately for each device.

8.3 Addressable Duct Type Smoke Detector (HVAC Equipment)

1. Addressable duct type smoke detector units shall be provided for monitoring of the supply and return air ducts. In the event of alarm from the duct detector, the related air conditioning equipment shall shut down automatically.
2. Addressable output modules shall be used to initiate shut down of the AHU.
3. The duct type smoke detector shall be mounted on the supply and return air stream of the ducting and designed to be able to lead to small amount of air from regular to high velocity air duct into a special chamber which contains of a smoke detector. The unit shall be able to operate reliably under the air velocity from 1m/sec to 20m/sec.
4. The unit shall house an addressable smoke detector as specified in

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Addressable Smoke Detector section above. The unit shall be suitable for connection directly to the Class A individual addressable loop.

5. The control function shall be performed via the addressable output module. Isolation buttons shall be provided at the control unit for isolation of the control function in case of system testing and maintenance.
6. A positive confirmation signal to confirm the intended control function has been carried out shall be provided at the control unit in plain text. The text shall describe type, designation and location of the AHU.
7. The "shut down" functions of air conditioning equipment shall be independent unless otherwise shown in the drawings or as directed by the Engineer.
8. The vendor shall provide the installation details of duct type smoke detector to the HVAC contractor and HVAC contractor will install the duct type smoke detector on ducts.

8.4 Flame Detectors

1. Flame detector shall use optics to see a small or a large fire far away without the heat or the smoke getting to it first. The optical flame detectors are programmed to trigger an alarm when they identify a flame through its wavelength.
2. The flame detector shall be microprocessor-based sensor technology.
3. The following primary types of optical fire-sensing technologies we can use:
   i. Infrared detectors (IR)
      It shall be with multi IR sensors to minimize false alarm.
   ii. Ultraviolet and Infrared (UV/IR)
      A dual-band flame detector is basically a combination of infrared sensors and Ultraviolet sensors sensitive to the IR and UV radiation emitted by a raging flame. The programmed detection mathematical algorithm shall be used to determine if an alarm is to be activated.
4. The optical flame detector shall be used for Silane BSGS, Ammonia BSGS, Acetylene BSGS, Solvent room, and Solvent waste room.
5. The flame detector output shall include fire alarm, fault, auxiliary contacts. It also has analog output capability with Hart communication.
6. The flame detector shall be at least certified SIL2 capable and ATEX Directive compliant.
7. The flame detector shall be RFI and EMC Directive compliant.
8. The flame detector shall have multiple sensitivity levels and calibrated automatic optical integrity.
9. The flame detector shall local status indicators.
10. The flame detector shall have event logging with time and date stamp.
11. The flame detector shall have integral wiring compartment for ease of installation.

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8.5 Addressable Manual Call Point

1. All manual call points shall be Pull Type or Break Glass, individual addressable type suitable for direct connection to the individual addressable loop. Glass of the manual call point shall be non-fragmental type. Conventional manual call point with interfacing unit will not be accepted. Manual call point shall be installed at 1.2 - 1.5 meters above finished floor level of respective area, it shall comply with local code.
2. The manual call point shall be RED in colour and consist of a built-in RED LED which flashes in case of activation.
3. It shall be possible to test for alarm at the manual call point without opening the call point cover or breaking the glass.
4. Manual call points shall comply with standard NFPA 72 or equivalent.
5. The addressable manual call points shall have a built-in line isolator which shall be able to isolate short circuits on the detector line such that any single short circuit in the detection loop will not impair the operation of the system or detectors.
6. The manual call point must be immune to external Electro-Magnetic Interference (EMI) generated by radio equipment. It shall have EMI protection up to 50 Volts/meter electric field strength tested in accordance to NFPA 72 and IEC 61000-4-3.
7. The manual call point shall be with plastic or polycarbonate cover. Outdoor manual call point shall be IP67.
8. Systems without built in isolator shall provide isolator separately for each device.

8.6 Addressable Monitor Module

1. Addressable monitor modules shall be used for monitoring of Sprinkler Flow Switches, Pressure Switches, Water Tank High/Low level, Pump Start/Stop status and other contact inputs.
2. It shall be able to connect directly along the individual addressable loop. An independent programmable software zone shall be provided for each interface unit.
3. The addressable monitor module shall be equipped with a built-in RED LED which shall be flashed in the event of activation.
4. It shall be possible to exchange the electronic parts without removing the housing or the wiring.
5. The addressable monitor module shall have a built-in line isolator which shall be able to isolate short circuits on the detector line such that any single short circuit in the detection loop will not impair the operation of the system or detectors.
6. The addressable monitor module shall be capable of operating in both dry and wet rooms.
7. The addressable monitor module must be immune to external Electro-Magnetic Interference (EMI) generated by radio equipment. It shall have EMI protection up to 50 Volts/meter electric field strength tested in accordance to IEC 61000-4-3.

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8.7 Addressable Control Module

1. The addressable control modules shall be designed to be placed along the detection line. The module shall provide a control output from the control unit to other equipment e.g. fire doors, smoke vents and AHU tripping.
2. The output contact of the module shall be rated to 240VAC/ 2A.
3. The addressable control module shall have a separate, fully supervised input to enable the control unit to obtain positive feedback to confirm that the output function has been carried out.
4. The addressable control module shall be disconnect-able by key code from the control unit.
5. The addressable control modules shall have a built-in line isolator which shall be able to isolate short circuits on the detector line such that any single short circuit in the detection loop will not impair the operation of the system or detectors.
6. The addressable control modules shall be capable of operating in both dry and wet rooms.
7. The addressable control module must be immune to external Electro-Magnetic Interference (EMI) generated by radio equipment. It shall have EMI protection up to 50 Volts/meter electric field strength tested in accordance to IEC61000-4-3.

8.8 Alarm Bell and Strobe Light

1. Alarm bells shall be electric bells suitable for 24VDC operation with continuous ringing tone of bell and should have sound level of at least 15 dB above the average ambient sound level or 5 dB above the maximum sound level having duration of at least 60 seconds. Refer to local code practice. The maximum sound level permitted in a space is 110dBA. If levels higher than this are required, special provisions such as visual alarm signal should be considered.
2. Strobe lights shall be provided in conjunction with alarm bells in areas where background noise level is high. The intensity of light from strobes shall be sufficient to draw the attention of people in vicinity.
3. Alarm bells shall be connected to the respective bell circuits which are monitored for open and short circuit fault. Alarm bells connecting to individual addressable lines are strictly prohibited.
4. The sounding of bell shall be on general alarm basis. However, provision shall be included for selective automatic bell sounding organization. (i.e. sounding of bells at the affected floor, two floors above and one floor below).
5. Sound boosters shall be provided in circuit wherever necessary.

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8.9 Fire Alarm Control Panel (FACP)

1. Fire Alarm Control Panel (FACP) shall contain microprocessor based Central Processing Unit (CPU) and power supply. The CPU shall communicate with and control the following used of equipment used to make up the system: intelligent addressable smoke and heat detectors, addressable modules, annunciators, and other controlled devices.
2. The system shall include a full featured operator interface control and annunciation panel that shall include a backlit LCD, individual color-coded system status LED, and an alphanumeric keypad with easy touch rubber keys for the field programming and control of the fire alarm system.
3. It shall be of modular design by means of plug-in cards to allow ease of maintenance and future system expansion minimum up to 4,000 addressable detecting points.
4. The control unit shall fully comply with the requirements of the standard NFPA 72 or equivalent.
5. The FACP shall have loop controllers with the following functions:
    i. Supervise and monitor all intelligent addressable detector and monitor modules connected to the system for normal, troubles and alarm conditions. Each loop shall be isolated and equipped to enunciate an Earth Fault condition.
    ii. Supervise all initiating signaling and notification circuits by way of connection to addressable monitor and control modules.
    iii. With its own microprocessor it shall be capable to operate local mode such that in the event of CPU failure, any addressable device input shall be capable of activating any or all addressable device outputs in the same loop.
6. All sensing devices such as detectors, manual call points, monitoring modules etc, shall be connected along the addressable loop regardless of the numbers of zones which shall be formed by programming at the FACP. Each loop shall provide, or capable of at least 300 intelligent / addressable devices. The number of individual addressable loops required shall be determined such that each loop shall have a spare capacity of 20% of the detector address and 20% of the module address.
7. Forming of zones within each loop shall be possible, according to the geographical and architectural requirements of the user’s premises.
8. The control unit shall provide this feature, in order to achieve a user-friendly system operation for any event being displayed. System which requires special programming equipment and setting of address for individual point at the sensing devices will not be accepted. It shall not require replacement of memory ICs to facilitate programming changes. All programming or editing of the existing program in the system shall be achieved without special equipment and without interrupting the alarm monitoring functions of the fire alarm control panel.
9. Minimum numbers of software programmable zones provided for each FACP must be corresponding to the drawings enclosed. Each software zone shall be able to provide a voltage free zone alarm output when required.

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10. All user data shall be stored in non-volatile memory to maintain the user data in case of total power failure or when the main microprocessor card is removed from the FACP.
11. All outputs for signaling or interlocking control functions shall be freely programmable.
12. The FACP shall contain software function for polling of fault diagnosis for easy fault finding.
13. When more than one FACP are required, the all control panels shall be able to perform communication each other and with one Main FACP via the 4 cores twisted Class A data communication loop. Each FACP must also be able to operate as a stand- alone unit in performing complete defined functions in case of the data communication with the Main FACP is interrupted.
14. The FACP shall be fully self-contained control unit with integrated emergency supply and networking capability that shall provide a link to a higher hierarchical system level (Main Fire Alarm Control Panel) in a communication network type system.
15. All status and information available in FACP shall be displayed on Main FACP.
16. The Main FACP shall be installed at the Fire Command Centre to monitor and control the entire fire alarm system. It shall allow control of all FACP and other devices connected along the fire alarm communication network. It should be able to handle up to 64 FACPs. The main FACP shall be able to monitor and process a minimum of 30,000 data points with additional 30% spare capacity for future system expansion.

8.10 Signal Display and Control Functions

1. The FACP shall provide a backlit alphanumeric LCD. It shall be able to display:
   i. Device status
   ii. Device type
   iii. Custom device label
   iv. View analog detector values
   v. Device zone assignments.
   vi. Analog Detector Sensitivity
   vii. All Program Parameters.
2. The FACP shall provide LED indicators for the following status:
   i. AC Power
   ii. Fire Alarm
   iii. Pre-Alarm
   iv. Security
   v. Supervisory
   vi. System Trouble
   vii. Other event
   viii. Signal Silenced
   ix. Point disabled
   x. Controls Active
   xi. CPU Failure

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3. The FACP display shall provide a Qwerty style keypad with control capability to command all system functions, entry of any alphabetic or numeric information, and field programming. At least two different password levels (Master and User) with minimum ten passwords shall be accessible through the display interface assembly to prevent unauthorized system control and programming. Only the master password shall allow access to password change screens.
4. The system programming shall be backed up via an upload/download program and stored on compatible removable media. A system back-up disk shall be completed and given in duplicate to Micron team upon completion of the final inspection. The program that performs this function shall be non-proprietary, in that, it shall be possible to forward it to Micron team.
5. The field programming and hardware shall be functionally tested on a computer against known parameters/norms which are established by the FACP manufacturer. A software program shall test Input-to-Output correlations, device type ID associations, point associations, time equations, etc. This test shall be performed on an IBM-compatible computer with a verification software package. A report shall be generated of the test result turned in to Micron engineer for record.
6. The FACP shall be able to provide the following software and hardware features:
    i. The system shall provide means to activate alarm signals to specific areas with an adjustable delay of the alarm after start of alarm processing. In addition, a positive alarm sequence selection shall be available that allows a delay (adjustable), all local and remote outputs shall automatically activate immediately.
    ii. To obtain early warning of potential fire alarm conditions, the system shall support a programmable option to determine system response to real-time detector sensing values above the programmed setting. At least two levels of Pre-alarm indication shall be available at the control panel: alert and action.

        Alert: It shall be possible to set individual smoke detectors for pre-programmed pre-alarm thresholds. If the individual threshold is reached. The pre-alarm condition shall be activated.

        Action: If programmed for Action and the detector reaches a level exceeding the pre-programmed level, the control panel shall indicate an action condition.
    iii. The system shall support a detector response time to meet annunciation requirement of less than 3 seconds.
    iv. It shall have means to turn off detector/module LED strobes for special areas.
    v. Smoke Detector sensitivity test meeting requirements of NFPA 72.
    vi. The system shall provide means to automatically initiate a reminder that troubles exist in the system.
    vii. Maintenance alert, with minimum two levels to warn excessive smoke detector dirt or dust accumulation.
    viii. The system shall provide at least nine sensitivity levels for alarm, selected by detector. The system also shall provide at least nine

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levels of Pre-alarm selected by detector to indicate impending alarms to maintenance personnel.

ix. The ability to display or print system reports.

x. Online or offline programming.

xi. History events shall include all alarms, troubles, operator actions, and programming entries. The panel shall maintain a history file of the last minimum 4000 events, each with a time and date stamp, and maintain minimum 1000 event alarm history buffer, which consist of the 1000 most recent alarm events. The history storage shall use non-volatile memory.

xii. The system shall provide means for all devices on any loop to be auto programmed into the system by specific address. The system shall recognize specific device type ID's and associate that ID with the corresponding address of the device.

xiii. The system shall provide means for setting Environmental Drift Compensation by device. When a detector accumulates dust in the chamber and reaches high level but yet still within the allowed limit, the control panel shall indicate a maintenance alert warning. When the detector accumulates dust in the chamber above the allowed limit, the control panel shall indicate a maintenance urgent warning.

xiv. The system shall provide means to enter at least 100 custom action messages.

xv. The software functions shall provide means to program a variety of output responses based on various initiating events. The control panel shall operate the program through lists of zones. A zone shall be listed when it is added to a point's zone map through point programming. Each input point such as detector, monitor module or panel circuit module shall support listing at least 10 zones into its programmed zone map.

xvi. The system shall support logic equations including but not limited to AND, OR NOT, and time delay equations to be used for advanced programming. When any logic equation becomes true, all output points mapped to the logic shall activate.

xvii. The CPU shall receive analog information from all intelligent detectors to be processed to determine whether normal, alarm, pre-alarm, or trouble conditions exist for each detector. The software shall automatically maintain the detector's desired sensitivity level by adjusting for the effects of environmental factors, including the accumulation of dust in each detector. The analog information shall also be used for automatic detector testing and for the automatic determination of detector maintenance requirements.

7. Mimic panels shall also be provided. It shall consist of metal housing with lockable door and silk screen floor plan with built-in LED for indicating the alarm status for the related area.
8. The mimic panel shall consist of a buzzer which sounds upon alarm activation from the related area with an alarm acknowledge button and lamp test button.
9. The signal to drive the corresponding LED of the mimic Panel shall be derived from the FACP via data transmission line.

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10. Overall system architecture shall be based on the principle of distributed processing that the FACP shall remain autonomous (independent) in evaluation, annunciation and control in case of the communication is interrupted between the FACP and the central fire alarm processing equipment.
11. All major equipment for the central fire alarm processing equipment shall be designed and manufactured by one and the same manufacturer to ensure system uniformity of standards, compatibility in operation, spare parts availability, trained support and competent maintenance services.

8.11 Central Fire Alarm Processing System

1. The Central Processing Unit shall contain and execute all control-by-event program including Boolean functions for specific action to be taken if an alarm condition is detected by the system. Such programs shall be held in non-volatile programmable memory, and shall bot be lost with system primary and secondary power failure.
2. The CPU shall also provide a real-time clock for time annotation, to the second, of all system events. The time-of-day and date shall not be lost if system primary and secondary power supplies fail.
3. The CPU shall be capable of being programmed on site without requiring the use of any external programing equipment. Systems that require the use of external programmers or change of EPROMs are not acceptable.

8.12 Fire Alarm System Communication Network

1. The fire alarm system communication network shall be a "free network" that messages is transmitted in a spontaneous manner so that the response time will not be affected by the size of the system and number of FACP connected.
2. Communication network shall be of class A loop transmission. Every path of the data transmission network must be monitored for open or short circuit fault.
3. In case of communication fault, fault signal shall be indicated at the affected FACP and the central fire alarm processing equipment at the Fire Command Centre. Such a fault shall not deteriorate the autonomous operation of the affected FACP.
4. Addressable device communication shall be Class A loop topology.
5. A single break along the data communication network shall indicate fault signal as describe in the above but the entire data transmission shall still be fully operational via the loop data transmission network.
6. In case of failure of a loop device, all other devices in the loop shall remain fully operational. The fault detector shall cause Trouble alarm at the FACP. When the failing device resumes correct operation or it is replaced and set with correct address, it shall automatically be re-inserted into the loop communication traffic after the alarm reset is done at the FACP.

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8.13 Graphic Workstation

1. The vendor shall be responsible to provide a proven system with software packages which enable the complete system operation in accordance with the requirements of the specification.
2. The Graphics Workstations shall be installed at the Fire Command Centre and Control room to monitor and control the entire fire alarm system. It shall allow control of all FACP and other devices connected along the fire alarm communication network.
3. The Graphics Workstation shall consist of CPU, Monitor, Keyboard and Mouse, Printer, latest Windows Operating system, and Graphic software.
4. The Graphics Workstation's fire alarm monitoring software shall be modular design which allows easy phase-by-phase extension of the system capacity and shall be able to handle up to at least 255 FACP.
5. The Graphics Workstation shall be able to monitor and process a minimum of 20,000 data points with additional 30% spare capacity for future system expansion.
6. The following minimum software modules shall be provided:
    - Operating program
    - System data base (process image)
    - Network monitoring and control
    - Display priority control
    - Command priority control
    - Output message generator
    - Peripheral drivers
    - Text/Graphic editors
    - "Help" facility
    - Service assistance
    - Diagnostic
    - Auxiliary programs
    - Access rights
    - License key
    - OPC support
7. The system software packages shall be specially made for the requirements of fire alarm system.
8. A comprehensive text, graphic and parameter editing program shall be provided for easy editing of user text, graphics and automatic system command functions respectively.
9. It shall be possible to assign different priority levels to the various types of system incidents or criteria. All system incidents shall be chronologically recorded in the system history file with date and time for subsequent reports. All logs in the system history file shall be retrievable according to the operator selection. Historical data override is not accepted, and alarm shall be generated the graphic workstation storage capacity is at high alarm. The following minimum parameters shall be made available for the operator to select:
    i. Types of system
    ii. Types of events
    Alarm, Fault, Active, zone off, zone Test, bypass/isolation alarm

and for All.

iii. Record on cause of alarm
To define cause of alarm actuation before reset of alarm message:
a. Genuine alarm
b. Deceptive alarm
c. Malfunction
d. Operator error
e. False response
f. Nuisance alarm
g. Not definable.

iv. Time of events

10. The software packages shall be able to perform predefined control by events and time dependent functions.
11. The system operating functions shall be password protected. At least ten different authorization levels for system operation shall be defined and shall be accessible via selected passwords.
12. The system editor functions shall only be accessible via assigned password. All system service functions shall be assigned to the highest authorization level. For each operator intervention, the identification number of the initiating password shall be logged by the system printer and recorded in the system history file.
13. The Graphics Workstation shall provide optimal conditions for successful intervention by means of rapid and simple processing and displaying of all necessary data in plain text or graphics. The procedures in operation shall be automatically guided and the information display shall not consist of any coded messages.
14. All messages shall be displayed on the system terminal in chronological order and the display shall fill up from top to bottom. An indication shall be displayed in the summary field when the display field is full to advise the operator that more messages are present in the memory.
15. Various types of message shall be displayed with different background color.
    i. Alarm message - RED
    ii. Fault message - YELLOW
    iii. Status message - WHITE
16. Detectors and/or actuators as referred to in the operation dialogue shall be indicated by their zone/area location. No technical code shall be used to identify system elements neither on the system terminal nor on the printer(s).
17. Un-acknowledged messages shall distinctively indicate to draw the attention of the operator. When an alarm message is reset, the display of that alarm shall be automatically deleted from the screen.
18. The alarm message per zone shall contain the following information:
    i. Time of day
    ii. Date
    iii. Type of alarm
    iv. Origin of alarm (Zone/Area of alarm indicated by area/building/floor/room)

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  v. Description of affected alarm location

19. The Graphics Station shall be capable of performing the following control functions:
    i. Acknowledge and reset alarm by either device, zone floor or building
    ii. Isolate by either device, zone, floor or building
    iii. Acknowledge any fault
    iv. Activate outputs of addressable control modules
20. All events related messages shall be displayed in plain text or graphics. No technical code shall be used for message display that the operator does not require technical knowledge for interpretation of messages displayed and operation of the system.
21. A keyboard shall be provided to form part of the system terminal for the following functions:
    i. Entry of password for protected operating functions.
    ii. Setting of time and date.
    iii. Assignment of password and operator access level.
    iv. Edit and modify system basic data, configuration parameters, program parameter etc.
    v. Edit and modify the plain user text and messages:
        a. Descriptions of FACP location
        b. Descriptions of alarm zone locations
22. The colour graphic function shall allow the interactive (real time) display of all alarm, fault and specific status events in the entire system. The events to be displayed shall provide clear identification of their origin by:
    i. Location of control units
    ii. Location of zones
    iii. Originating source
        a. Automatic detector
        b. Manual call point
        c. Extinguishing System
23. The graphics presentation shall have a minimum of four hierarchical levels:
    i. Area overview
    ii. Building overview
    iii. Floor overview
    iv. Room overview
24. Picture generation shall be effected by means of a versatile interactive picture editing program. No specific training shall be required for picture generation.
25. The system shall have the capacity to store minimum of 800 pictures for processing and displaying.

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8.14 Master Mimic Panel

1. A Master Mimic Panel with the following LED indicators and control buttons shall be provided. The coding /decoding unit for Master Mimic Panel (Mimic Driver) shall be able to supply the required signals suitable to drive the LED indicators and perform control functions as specified hereunder:
   i. Indications:
      a. General Alarm Status from each building
      b. Alarm from each Sprinkler Flow Switch
   ii. Control Function: Lamp Test
2. The Master mimic Panel shall be of Perspex material, with sectional view of the building silkscreen printed on reversed side. LEDs and switches shall be mounted on the Perspex board to indicate and control the above status. The Perspex board shall have a minimum of 6 colors for clarity. The master mimic Panel shall be mounted onto the Master Alarm Panel.
3. The coding and decoding unit for the master mimic board shall be of microprocessor based with sufficient outputs and provided with 30% spare capacity. The master mimic Panel shall display all signals as described in the above on a sectional view of the building. The final graphic arrangement of the master mimic Panel shall be submitted to Authority representative and Micron for approval before fabrication.

8.15 Power Supply

1. The fire alarm control panel requires dual incoming power supplies with battery/UPS-backup emergency power (essential).
2. Each power supply shall contain suitable over-voltage protection to prevent any malfunction or damage due to power line surges.
3. Each power supply unit shall be equipped with standby battery of nominal voltage of 24VDC with the capacity to maintain its operation for at least 24 hours after mains failure and shall be sized to comply with local code. All circuits shall be powered-limited per UL 864 requirements.
4. In the event of power failure, the power supply unit shall automatically switch over to battery power without a break to maintain the operational condition of the system.
5. The standby batteries shall automatically be in charged condition by a built-in short circuit proof charger.
6. The load shall automatically be switched off when the voltage drops to protect battery cells from being damaged by complete discharge.
7. Gas-tight maintenance free lead-calcium batteries shall be used for standby power supply and shall have a minimum life cycle of five (5) years.
8. Each power supply unit and charger circuit including all fuses shall be supervised. Any malfunction or blown or missing fuse shall result in a fault indication on the affected unit and the FACP.
9. When the AC power is restored, the power unit shall automatically

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revert to normal operation without any manual restoring procedure.

10. The power supply for all field devices is addressable as part of the device communication loop which linked up to FACP. It shall supervise for battery charging failure, AC power loss, power brownout, battery failure, and ground fault detection. In the event of trouble condition, the addressable power supply shall report the incident and the applicable address to the FACP via communication loop.
11. The power supply mounts in either the FACP or its own panel.

8.16 Fire Alarm Wire and Cable

Cables insulated with general purpose PVC shall not be installed in any situation where the sustained ambient air temperature is liable to exceed 60 degrees C for long periods. If such situations cannot be avoided, cables having heat resistant PVC insulation shall be used complying with local code requirement.

Suitable additional protection for the cable shall be provided at any point where they are likely to be subjected to mechanical damage.

PVC insulated non-sheathed cable shall be laid in metal conduit or metal trunking complying to local code, exclusive to the fire alarm system. Rigid PVC conduits and fittings may be used in situation where the ambient temperature is below 60 degrees C. Suitable additional protection for the conduits shall be provided at any point where they are likely to be subjected to mechanical damage.

If a common duct or trunking is to be used to contain both fire alarm circuits and those of any other services, the fire alarm circuits shall be wired in fire resistant cable. Alternatively they may be wired in PVC insulated cable provided they are separated from the cables of other services by a rigid and continuous partition of non-combustible material affording them complete enclosure when the covers of the duct or trunking are in place.

Cables laid underground shall be run in ducts, PVC insulated and sheathed cable conforming to the requirement of IEC60227-4 shall be used. Cables laid direct in the ground shall be PVC insulated and sheathed, armored and sheathed overall.

Any telephone-type cable should be allowed for use the wiring between the fire alarm panel and the repeater or mimic panel for secondary indication and shall be protected against mechanical damage by the use of conduit or trunking.

In selecting conductor sizes, regard should be paid to physical strength and to limitations imposed by voltage drop. Voltage drop across a cable should not be such as to prevent devices from operating within their specification limits, even under minimum supply and maximum load conditions. Consideration should be given to any possible extensions to the system.

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9.0 Interfacing and Communication

9.1 Interfacing for Alarm and Interlock

1. Interface with Access Control System

   Fire alarm system shall be interfaced with building's access control system for release of access-controlled doors (doors with card readers) in case of detection of fire in any of the zones. Enough quantity of addressable output modules shall be provided for this purpose.
2. Interface with Fire Shutter Door and Automatic Release Fire Door.

   Addressable output modules shall be provided for this function.
3. Smoke Extraction and Pressurization Fan

   Provision shall be made in the main alarm panel to switch off the smoke extraction and pressurization fan through MCC starter unit via FMCS.
4. Monitoring of Fire Sprinkler Pumps and Water Storage Tanks Status.

   The fire alarm system shall monitor status of each fire pump and storage tanks. The addressable input modules shall be used for this function.
5. Status monitoring of each fire pump shall include:
   - Mains Power Supply failure
   - Pump Running
   - Pump Tripped.

   The following status shall be monitored for each Diesel storage tank:
   - Water Tank High Level
   - Water Tank Low Level
6. Monitoring Of Sprinkler Subsidiary Valves
   - Each sprinkler subsidiary valve shall be individually monitored for closed status.
   - Addressable input modules shall be used for this function.
7. Outputs shall be freely programmable to provide the required signal for automatic actuation of Public Address System.
8. Lift

   The Fire Alarm System has to be interfaced with all the Lifts in the building. Lift Recall / Homing shall occur in case of Fire Alarm System activation in the building.
9. Gas Based Extinguishing System.

   The activation of the system is made automatically by a full confirmed fire alarm or manually by pushing emergency button located at an easily accessible place adjacent to the area under protection.
10. Carpark Barrier

    Carpark barrier shall be release when fire alarm is activated.

9.2 Interfacing with Facility Monitoring and Control System (FMCS)

1. Voltage free normally open contact output shall be provided for the following signals for interfacing with FMCS:
   i. Master Alarm Signal

ii. Master Fault Signal
iii. Common Alarm signal from automatic detectors
iv. All Fire pumps and Water Tanks monitoring signals

The interfacing signal contacts shall be freely programmable

2. The vendor shall liaise with the FMCS contractor / vendor to ensure satisfactory interfacing between the two systems.

9.3 IT Requirement

1. The Graphic workstation will be coordinated with IT for integration and security.
2. Control interlock and monitoring should be dedicated and separate network.

10.0 Execution

10.1 Installation

1. Installation shall be in accordance with the NEC, NFPA 72, local code, and as recommended by the equipment manufacturer.
2. All conduit, junction boxes, conduit supports and hangers shall be concealed in finished areas and may be exposed in unfinished areas. Smoke detectors shall not be installed prior to the system programming and test period. If construction is ongoing during this period, measures shall be taken to protect smoke detectors from contamination and physical damage.
3. All fire detection and alarm system devices, control panels and alarm devices shall be flush mounted when located in finished areas and may be surface mounted when located in unfinished areas.
4. Mount outlet box for electric door holder to withstand 80 pounds pulling force.
5. Connections to Other Systems: Make conduit and wiring connections to door-release devices, sprinkler-flow switches, sprinkler valve tamper switches, fire suppression system control panels, HVAC motor starter circuits, electrical shunt trip circuit breakers, and elevator control panels.
6. Equipment shall be installed in adequate ventilation which is permitted for the equipment operation.

10.2 Field Quality Control

The service of a competent, factory-trained engineer or technician authorized by manufacturer of the fire alarm equipment shall be provided to technically supervise and participate during all of the adjustments and tests for the fire alarm system. All testing shall be in accordance with NFPA 72 or equivalent.

1. Before energizing the cables and wires, check for correct connections and test for short circuits, ground faults, continuity, and insulation.
2. Close each sprinkler system flow valve and verify proper supervisory

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alarm at the FACP.

3. Verify activation of all water flow switch.
4. Open initiating device circuits and verify that the trouble signal actuates.
5. Open and short signaling line circuits and verify that the trouble signal actuates.
6. Open and short notification appliance circuits and verify that trouble signal actuates.
7. Ground all circuits and verify response of trouble signals.
8. Check presence and audibility of tone at all alarm notification devices.
9. Check installation, supervision, and operation of all intelligent smoke detectors using the walk test.
10. Each of the alarm conditions that the system is required to detect should be introduced on the system. Verify the proper receipt and proper processing of the signal at the FACP and the correct activation of the control points.
11. Functional test shall be according to sequence of operation including cause and effect matrix.
12. The original inspection/test record and completion documentation, including certificate of completion shall be submitted to Micron for acceptance.

10.3 Training

Provide the Owner a minimum of 8 hours of instruction in the operation of the system, test equipment, and necessary maintenance of the detection devices.

11.0 HIGH SENSITIVITY SMOKE DETECTION (HSSD)

This section includes the requirements necessary to furnish and install a complete High Sensitivity Smoke Detection (HSSD). The HSSD system shall be installed in cleanroom areas. The HSSD can also be installed in other areas.

11.1 System Description

1. The HSSD system shall consist of a highly sensitive, short wavelength laser-based, particle imaging and light scattering smoke detector, aspirator, and filter. The system shall be modular, with each detector having a display with indicator LEDs and a reset control button and/or optionally with LCD Display showing detector status including fault categories and smoke level.
2. The HSSD system consists of an air sampling pipe distribution network which draws an air sample which is monitored for smoke conditions at the detector unit. Upon detection of incipient smoke, the HSSD system indicates an alarm condition.
3. The HSSD system shall support equipment which include graphic workstation and high-level interface with building fire alarm system.
4. The HSSD system shall provide four output levels corresponding to

Alert, Action, Fire 1 and Fire 2. These levels shall be programmable and able to set at sensitivities ranging from 0.005-20% obs/m (0.0016-6.25% obs/ft) with a resolution of 0.0002% obs/m (0.00006% obs/ft).

5. The system shall report any fault on the detector by using configurable fault relay outputs, via a peer-to-peer network or by communication to a monitoring software tool running on a PC.
6. The system shall be self-monitoring for filter contamination and incorporate a flow sensor in each pipe inlet and provide staged airflow faults against flow fault thresholds that may be determined and set for each pipe individually.
7. The HSSD system to include wiring, raceways, pull boxes, terminal cabinets, mounting boxes, control equipment, detectors, air sampling pipe network, power supply with backup rechargeable batteries, accessories, and other miscellaneous items required for a complete functional system providing continuous detection of the desired area.

11.2 Design Criteria

1. The HSSD system design, installation, commissioning, and testing is accordance with applicable Codes, this Section, the manufacturer’s guidelines, and requirements of the local authority.
2. HSSD consists of aspirating smoke detectors and air-sampling pipe distribution network which includes piping, tubing, fittings, and fasteners. The piping originates from each detector and extends to the protected area(s). Air sampling points are placed along the air sampling pipe distribution network. The locations and diameter of the air sampling points in the protected area(s) are supported with the manufacturer’s computer-based design modeling tool.
3. HSSD system to include interface and connections to the building fire alarm.
   i. Configurable alarm and trouble relays outputs transmit signals to building fire alarm system and other systems.
   ii. Where applicable, a high-level interface may communicate with building fire alarm system or FMCS workstation.
4. Each detector incorporates an integral repeater and loop isolator for communications on a multi-point serial communication network.
5. Alarm Response: Smoke alarm response to be less than 90 seconds from the moment of smoke injection at furthest sample point to alarm condition reported at controller unit.
6. Continuity: Sampling of air to be continuous for detector sensing. Air flow faults to be indicated within 60 seconds when the airflow is outside the manufacturer's specified range.
7. Event Logging: System to automatically record faults, smoke alarms, and setpoint adjustments into an internal buffer.
8. Threshold Settings: Adjustable threshold settings to be made for day, night, weekend, and holiday periods.
9. Sample Pipe Network:
   i. Layout of the air sampling pipe network to provide area coverage per NFPA72 and per manufacturer’s engineering calculations,

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isometric air-sampling drawings, and manufacturer's installation instructions.

ii. Layout of the sampling pipe network to provide for a cross-sectional sampling of the return airstream. Careful consideration to be given to National Fire Protection Association (NFPA) 72 guidelines to the number of sampling points and spacing with regard to high ceilings and forced-air ventilation applications.

iii. Sample piping length, diameter, material of construction, sample hole size, and location to be in accordance with manufacturer's engineering calculations, isometric drawings, installation instructions and to be sized to meet the intended detection area.

iv. Detector shall exhaust into an equal or lessor pressure and in no case shall the Δp between detector sampling pipes and between sampling pipes and detector exhaust exceed 100 Pa (0.4 inches of water).

10. For detection of smoke concentration within ceiling void(s), where required by local authority.
    i. Provide appropriate number of detectors to efficiently and adequately cover the protected area(s).
    ii. Air sampling pipe distribution networks is constructed of rigid pipe mounted to the underside of the structural ceiling.
    iii. Sampling points are drilled directly into the air sampling pipe distribution network. Air sampling point is oriented downward and be within four inches below the underside of the structural ceiling.
    iv. For structural ceilings that have beams, girders, solid joist or a waffle-like construction, etc., and that meet the special provisions of NFPA-72, air sampling pipe distribution networks shall mount to the underside of structural beams with stanchions extending up into each beam pocket at locations where sampling points are required.
    v. Where the ceiling void is used as an air plenum, sampling points shall be oriented downward 20 degrees to 45 degrees towards the incoming airflow.
    vi. Maximum coverage per sampling point shall not exceed 400 sq ft.
    vii. Sampling points shall be installed in an area on the ceiling that is free from obstructions for a minimum of 18 inches on all sides.
    viii. Placement of sampling points nearest to walls shall be greater than 12 inches and not exceed 7ft.
11. For detection of smoke concentration within open areas, where required by local authority requirement.
    i. Provide appropriate number of detectors to efficiently and adequately cover the protected area(s).
    ii. Air sampling pipe distribution networks shall be constructed using rigid pipe mounted to the underside of the structural ceiling.
    iii. Sampling points shall be drilled directly into the air sampling pipe distribution network, be oriented downward and be within 4 inches below the underside of the structural ceiling.
    iv. For structural ceilings that have beams, girders, solid joist or a waffle-like construction, etc., and that meet the special provisions of NFPA-72, air sampling pipe distribution networks shall mount to the underside of structural beams with stanchions having sampling

point extend up into each beam pocket at locations where sampling points are required.

v. Where accessible ceilings are present sampling points shall be remotely mounted to the underside of the accessible ceiling, be oriented downward and be within 4 inches below the underside of the accessible ceiling. Remote sampling points shall interconnect with air sampling pipe distribution network located above the accessible ceiling via use of flexible "capillary" tubing.

vi. For efficiency, air sampling pipe distribution networks located above the accessible ceiling shall have "capillary" tubing running either side of each sampling pipe run to the corresponding remote sampling point.

vii. Maximum coverage per sampling point shall not exceed 400 sq ft exception being where obstructions such as air supply diffusers would otherwise prevent this spacing objective from being met, in which case placement shall be just outside perimeter of obstruction.

viii. Sampling points shall be a minimum of 3ft away from air supply diffusers.

ix. Sampling points shall be installed in an area on the ceiling that is free from obstructions for a minimum of 18 inches on all sides.

x. Placement of sampling points nearest to walls shall be greater than 12 inches and not exceed 7ft.

12. For detection of smoke concentration within accessible floors, where required to meet local authority requirements:

    i. Provide appropriate number of detectors to efficiently and adequately cover the protected area(s).

    ii. Air sampling pipe distribution networks shall be constructed using rigid pipe mounted and secured to raised floor support posts as close to the underside of floor tiles as possible.

    iii. Sampling points shall be drilled directly into the air sampling pipe distribution network and be oriented downward towards the slab.

    iv. Where the floor void is used as an air plenum, sampling points shall be oriented downward 20 degrees to 45 degrees towards the incoming airflow.

    v. Maximum coverage per sampling point shall not exceed 200 sq ft.

    vi. Sampling points shall be free from obstructions for a minimum of 18 inches on all sides.

    vii. Placement of sampling points nearest to perimeter walls extending to the slab shall be greater than 12 inches and not exceed 7ft.

13. For detection of smoke concentration within contained hot aisles:

    i. Provide appropriate number of detectors to efficiently and adequately cover the protected area(s).

    ii. Air sampling pipe distribution networks shall be constructed using rigid pipe mounted at a point just before mechanical forced air leaves the contained hot aisle.

    iii. Sampling points shall be drilled directly into the air sampling pipe distribution network and be oriented 20 degrees to 40 degrees downward towards the incoming airflow.

    iv. Sampling point spacing shall be based upon sampling point distribution of minimum one sampling point per 6ft (1.8m) of

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linear distance, not to exceed 36 square feet per sampling point.

v. Each hot aisle shall be served by an individual air sampling pipe zone or sector (eg. four pipes on a four-zone apparatus with each pipe serving an individual hot aisle).

14. For detection of smoke concentration at return air paths:
    i. Provide appropriate number of detectors to efficiently and adequately cover the protected area(s).
    ii. Sampling at return air grilles:
        a. Air sampling pipe distribution networks shall be constructed using rigid pipe mounted to face of return air grilles prior to filtration in accordance with Manufacturer's guidelines, and this Specification.
        b. Sampling points shall be drilled directly into the air sampling pipe distribution network, be oriented downward 20 degrees to 45 degrees towards the incoming airflow and shall not be placed outside of grille area.
        c. Sample point spacing for return air grille detection shall be based upon sampling point distribution of a minimum of one sample point per every four-square feet of grille area.
        d. Sampling points covering return air grilles shall standoff 3 to 6 inches from the grille in order to avoid the low or negative pressure point directly at the grille surface.
        e. A means to disconnect the sampling pipe from return air grilles shall be provided and based on Manufacturer's guidelines.
        f. Mechanical standoff fasteners used to support the sampling pipe network shall be fit for purpose. Plastic zip ties are not permitted.
    iii. Sampling within return air chase:
        a. A detector serving a return air chase shall be dedicated to the mechanical air handler unit's return air chase in which it is monitoring.
        b. Air sampling pipe distribution networks shall be constructed using rigid pipe with sampling to occur within return air chase.
        c. The exhaust port of each detector shall be returned within the return air chase in which it is monitoring so as to maintain an equal pressure across air sampling point inlet and exhaust port.
        d. Sampling port spacing within return air chase shall be spaced in accordance with Manufacturer's guidelines and be oriented 20 degrees to 45 degrees towards the incoming airflow.
        e. Application shall be in accordance with the Manufacturer's published induct sampling guidelines.
15. Manufacturer's specific software modeling tool which uses air sampling pipe distribution network parameters shall be used to predict performance of pipe network design(s). Each of the following performance requirements shall be met:
    i. Maximum Fire 1 sensitivity at each sample point for detector shall not exceed 1 percent obs/ft (1.5 percent obs/m) and shall be within 30 percent of this target.
    ii. Smoke transport time shall not exceed 90 seconds measured from the furthest sampling point (excluding Benchmark Test Point) on

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each pipe run back to the detector. A minimum 5 percent buffer shall be maintained to account for field variations that may occur but in no case shall exceed the 85 seconds objective.

16. Where local codes and standards dictate placement and spacing of sampling points that is more stringent than those specified herein, local codes and standards shall prevail.
17. All sampling points shall be marked with the Manufacturer's standard identification labels as specified herein.
18. Detector shall be installed in an accessible location and at an accessible height not to exceed 6ft as indicated in this Specification.
19. A Benchmark Test Point shall be provided at the furthest end of each pipe run, opposite end of the detector. Provisions shall be made so that this test point can be located maximum 6 feet above finished floor. The test point shall be constructed using a normally closed industrial quick connect fitting with an orifice of nominal 1/8 inch when open to facilitate benchmark performance verification as indicated in this Specification. This remote test point is intended to benchmark system performance at time of initial commissioning and during routine test and inspection. The test point shall be labeled documenting benchmark system performance at time of commissioning using Manufacturer supplied labels intended for this purpose. Benchmark labels shall be placed just above test points and be positioned so that they are visible without obstruction.
20. Installation and materials shall be in accordance with Manufacturer's guidelines.
21. System Power Supply and Batteries:
    i. Power supply is to be adequate to serve control panel and remote detectors, including battery-operated emergency power supply with capacity for operating system in standby mode for 4 hours followed by alarm mode for 5 minutes.
    ii. Batteries to be sized for operating the system in standby mode for 24 hours in the absence of an automatic-starting engine-driven generator.
22. Equipment Approvals/Listing:
    i. Factory Mutual (FM)-approved.
    ii. Underwriters Laboratories Inc. (UL)-listed.

11.3 Submittal Requirement

1. Manufacturer's catalog data, to include material description, agency approval, and listing.
2. Submittal Schedule shall include, but not limited to:
    i. Engineering calculations that include the maximum smoke transport time from the last sampling point and airflow balance details from each sampling point.
    ii. Installation drawings showing details of the sampling pipe network proposed layout, piping isometrics, and sampling hole sizes and locations.
    iii. After approval of shop drawings, provide final calculations.

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iv. Manufacturer's letter stating that system is operational.
v. Manufacturer's certificate stating that system meets or exceeds specified requirements.
vi. Completed system checkout field report with copies of system response tests.

3. Shop Drawings:
   i. Provide mounting details of detectors, displays, programmers, power supplies and other boxes to building structure, showing fastener type, sizes, material, and embedded depth where applicable.
   ii. Air sampling pipe, fittings and pipe details including pipe routing and support.
   iii. The following Calculations and Wiring Diagrams shall also be included on the shop drawings:
      a. Provide calculations for the battery stand-by power supply taking into consideration the power requirements of all auxiliary components under full load conditions. Include battery size calculations.
      b. Show system components, including detectors and cabinets, locations, quantities, and full schematic of system wiring showing conductor size, routings, quantities and connection details.
   iv. Reflected Ceiling Plans: Show ceiling penetrations, ceiling-mounted items, and method of attaching hangers to building structure.
   v. System Performance Calculations: Submit report generated from Manufacturer’s system performance calculation program illustrating performance criteria such as transport time to the detector. The system transport time shall not exceed 90 seconds from the furthest sampling point back to the detector.
   vi. A complete sequence of operation.
4. Permit Approved Drawings: Working plans, prepared according to local codes and standards, that have been approved by Authorities Having Jurisdiction. Include design calculations.
5. Maintenance data
6. Warranty information

11.4 Quality Assurance

1. Provide the Work in accordance with NFPA 70 (National Electric Code NEC) or equivalent. Where required by authority, material and equipment shall be listed and labeled by a nationally recognized testing laboratory or other organization acceptable to the authority representative, in order to provide a basis for approval under the above listed agency.
2. Materials and equipment manufactured within the scope of standards published by UL shall conform to those standards and shall have an applied UL listing mark or label.
3. Provide materials and equipment acceptable to Authority for Class, Division, and Group of hazardous area indicated.

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4. Material and equipment must be listed and labeled for its intended purpose, environment, or application, especially when used in extreme climate areas.
5. Contractor Qualifications:
   i. All contractors involved with the design and installation of HSSD systems shall be very experienced with the systems that they are designing / installing per the occupancy use. A minimum of 5 years experience in the installation of HSSD systems on staff to supervise the design and installation is required.
   ii. The installing contractor shall have successfully passed advanced certification training and be listed by the HSSD system Manufacturer as an accredited contractor, trained and certified to model, design, install, program, test and maintain the HSSD system and shall be able to produce a certificate stating such upon request.
6. Testing Technicians Qualifications:
   i. Testing technicians of HSSD systems shall be trained and qualified by the HSSD system Manufacturer in the proper operation of the system per the occupancy use.
   ii. System testing technicians or others conducting system programming, certification, power up, and system commissioning shall have had advanced certification training, be listed by the HSSD system Manufacturer as an accredited contractor and shall be able to produce a certificate stating such upon request.
   iii. Testing technicians shall be an accredited fire alarm technician to perform such work.

11.5 Components

11.5.1 Detector

i. The detector, Filter, Aspirator, and Relay Outputs shall be housed and shall be arranged in such a way that air is drawn from the fire risk area by an aspirator and a sample passed through a sample filter and detection chamber.
ii. The detector shall employ a short wavelength laser light source and incorporate particle imaging and light scattering using a two-dimensional image sensing array and scatter pattern measurement using photodiodes.
iii. The detector shall have an obscuration sensitivity range of 0.005-20% obs/m (0.0016-6.25% obs/ft) with resolution of 0.0002% obs/m (0.00006% obs/ft).
iv. The detector shall have four independent field programmable smoke alarm thresholds across its sensitivity range with adjustable time delays for each threshold between 0-60 seconds.
v. The detector shall employ modular construction allowing field replacement of the filter, chamber and aspirator.
vi. The detector shall allow future hardware expansion via stackable modules placed either on top or below the detector.
vii. The detector shall incorporate facilities to transmit the following fault categories: Detector fault, Air flow fault, Filter

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fault, system fault, Zone fault, Network fault, Power fault, Chamber fault, Module fault.

viii. The detector shall support the generation and transmission of urgent and minor faults. Minor faults shall be considered as servicing or maintenance signals. Urgent faults indicate the unit may not be able to detect smoke.

ix. The filter shall be a disposable filter cartridge and shall be capable of filtering particles in excess of 20 microns from the air sample.

x. A second filter shall be ultrafine, removing more than 99% of contaminant particles of 0.3 microns or larger, to provide a clean air barrier around the detector's optics to prevent contamination and increased service life.

xi. The aspirator shall be a purpose-designed impeller air pump with applicable transport time as per local code.

xii. The Assembly must contain seven (7) or more relays for alarm and fault conditions. The relays shall be software programmable to the required functions. The relays must be rated 2 Amp at 30Vdc.

xiii. The detector shall have built-in event and smoke logging. It shall store smoke levels, alarm conditions, operator actions and faults. The date and time of each event shall be recorded. Each detector zone shall allow storage of up to 20,000 events and does not require the presence of a display in order to do so.

xiv. The detector shall incorporate a monitored voltage-free input, to be used with isolated relay contacts, which is supervised.

xv. The detector shall provide LED user interface with a button to support Reset and Disable commands:

a. Four LEDs are to indicate Alert, Action, Fire 1 and Fire 2 alarm events
b. One trouble LED
c. One disable/standby LED
d. Power On/Off indication.

11.5.2 Power Supply

HSSD power supplies:

i. Power supply to be by the same manufacturer as the HSSD Controller unit it is supplying.
ii. Input voltage 120Vac or 240 Vac shall be supervised.
iii. To contain 4-hour battery backup and use an additional battery cabinet, if required.
iv. Power supply and batteries to be mounted in their own lockable enclosure (self-contained).
v. Output voltage to detector units to be 24 Vdc power limited.
vi. To contain fault reporting of a ground fault and/or alternating current (AC) input loss, low AC input voltage, short circuit of battery leads and short circuit of any direct current (DC) output.
vii. Not to exceed the output limitations as recommended by the manufacturer.

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11.5.3 Air Sampling Pipe, Capillary Tubing, Remote Sampling Points, Fittings and Mounting Hardware

i. Material for sampling pipe, capillary tubing, remote sampling points and associated fittings shall be in accordance with Manufacturer's guidelines, local codes and standards.
ii. Pipe and tubing inside diameter shall be in accordance with Manufacturer's guidelines and as specified in air sampling pipe distribution software modeling tool.
iii. Pipe and tubing interior shall be smooth bore.
iv. Pipe and tubing shall be marked with the listing information and marking requirements as per local codes and standards.
v. All fittings shall be made with compatible pre-formed elbows, tees, couplings, end caps and unions.
vi. All joints in the sampling pipe must be air tight and made by using compatible solvent cement, except at entry to the detector.
vii. Mechanical pipe fasteners and hangers shall be approved for use with the pipe material in which it is supporting. Fasteners and hangars shall allow pipe to freely slide in and out to facilitate expansion and contraction of the material. The sampling pipe to be constructed with manufacturer-approved pipe and fittings.

11.5.4 NETWORKED SYSTEMS

i. Aspirating smoke detectors, display modules, or programming units are linked on a daisy-chained loop. This allows remote configuration, monitoring, control, and programming.
ii. Network Communication: two-wire fault tolerant closed-loop system. Network units polled sequentially for alarms, faults, or trouble conditions.
iii. The networks shall be host by computer workstations.

11.5.5 COMPUTER WORKSTATION

Aspirating smoke detection monitoring allows remote configuration and monitoring from central location via communication loop. The Workstations shall be located in FFC room and Control room.

11.5.6 SOFTWARE

i. The latest versions of system configuration, testing and modeling software to be provided to Owner.
ii. Required software and communication hardware for communication, configuration and testing to the building fire alarm system.
iii. Required hardware devices for communication, configuration and testing of system equipment to be provided to Owner.

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11.6 Installation

1. Comply with local codes and standards, and in accordance with the Manufacturer's written instructions for installation of HSSD systems.
2. Air Sampling Smoke Detector
   i. Detector shall be installed in accordance with this Specification and the Manufacturer's written installation and instruction manuals.
   ii. Detector assembly shall be mounted to a wall in the protected room at a height of approximately 48-60 inches or 1.2-1.5m to top of detector measured above finished floor.
   iii. Mounting location shall be a fully accessible and visible location.
   iv. Mounting or attachment to site equipment, cable trays, movable walls, other equipment or equipment supports is not permitted without express written approval.
   v. Piping network insertion into the detector inlet shall not be glued.
   vi. Flexible tubing for termination of the sampling pipe network into detector inlet is not permitted.
   vii. In seismic zones installation techniques shall be in compliance with the geographical seismic zone requirements per local authority requirement.
3. Pipe Mounting
   i. The air sampling pipe network shall be mounted in accordance with the Manufacturer's written installation and instruction manuals, and local codes and standards. The hardware used for mounting will depend upon the design and site requirements.
   ii. To minimize flexing the pipes shall be secured every 5ft or 1.5m.
   iii. When installing a pipe network in areas subject to high temperature fluctuations allow for the contraction and expansion of pipes.
   iv. Where expansion or contraction of pipes is likely either after installation or on a continuous basis, do not place pipe clips adjacent to couplings and socket unions as these may interfere with the movement of the pipe.
   v. No bends are permitted within the first 20 inches or 500mm from the detector inlet.
   vi. Piping shall be mounted as close to the ceiling as possible.
   vii. The routing of the piping network shall be coordinated with potential obstructions, including cable trays, grounding bars, and HVAC ductwork.
   viii. All changes in direction shall be made with standard elbows or tees.
   ix. All joints shall be air-tight and made by using solvent cement, except at the entry to the detector assembly.
   x. Flexible tubing for termination of the sampling pipe network into detector inlet is not permitted.
   xi. All pipes shall be supported by mechanical hangers attached to the structure of the building. Not more than one foot of pipe shall extend beyond the last hanger of each sampling pipe. The final installation shall result in no noticeable deflection in the piping network.
   xii. Attachment of air sampling pipe networks to cable trays is prohibited.

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xiii. Air sampling pipe network shall be labeled with the detector Manufacturer's standard identification marking at least 20ft or 6m intervals.

xiv. Placement of the air sampling pipe network shall take into consideration appropriate sampling point locations and spacing.

xv. All air sampling pipe network penetrations of fire-rated partitions shall be sealed with a listed and/or approved through-penetration firestop material.

xvi. Swarf and other debris accumulated in air sampling pipe network during installation and assembly shall be removed by negative airflow prior to connection to the detector inlet(s).

4. Capillary Sampling Network:
   i. A capillary tube is a length of tubing connected to the main sampling pipe with a sampling hole at the end. The purpose of these tubes is to extend the sampled area from the main pipe network.
   ii. Capillary tubes are used when sampling an enclosed areas, such as cabinets or above ceiling.
   iii. The maximum capillary tube internal diameter should be 9.5 mm or 0.375 inch.
   iv. For above ceiling areas, remote sampling point locations shall align in center of acoustical ceiling tiles where possible provided spacing. Appearance shall be uniform in terms of sampling point row alignment. Coordinate layout with T-Bar grid.
   v. The maximum length of the capillary tube shall be 8 m or 26ft.
   vi. An additional 2ft of capillary tubing shall be provided for each drop to permit future relocation of remote sampling points (not to exceed 26ft limitation).
   vii. Remote sampling points shall be labeled with the Manufacturer's standard identification marking.
5. Locate and install the detector assembly for optimum response time.
6. Install the sampling pipe network in accordance with the manufacturer's recommendations. Make necessary adjustments for optimal air sampling.
7. Coordinate the mounting and penetration of sampling pipe with mechanical equipment installation and structural elements.
8. Clean exterior of exposed sampling pipe of dirt and markings.

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11.7 Initial Commissioning Settings

1. Initial Commissioning Settings for HSSD system.

   Alarm threshold settings for air sampling smoke detection apparatuses shall be configured to achieve target port sensitivities above recorded average ambient peak level of 0.20% obs/ft (0.6% obs/m) for Alert, 0.6% obs/ft (2% obs/m) for Action, 1.0% obs/ft (3.3% obs/m) for Fire 1, and 2.5% obs/ft (8% obs/m) for Fire 2. Data review and analysis shall consider the normal base line or ambient pollutant level recorded, as well as all deviations from the established base line as recorded by the apparatuses event log. The recorded data shall then be used in conjunction with the following formulas to calculate alarm thresholds:

   i. Fire 2 – Divide target sampling port sensitivity of 2.5% obs/ft (8% obs/m) by number of holes in overall system pipe network then add the recorded average ambient peak level to derive at the Fire 2 alarm threshold.
   ii. Fire 1 – Divide target sampling port sensitivity of 1.0% obs/ft (3.3% obs/m) by number of holes in overall system pipe network then add the recorded average ambient peak level to derive at the Fire 1 alarm threshold.
   iii. Action – Divide target sampling port sensitivity of 0.6% obs/ft (2% obs/m) by number of holes in overall system pipe network then add the recorded average ambient peak level to derive at the Action threshold.
   iv. Alert – Divide target sampling port sensitivity of 0.20% obs/ft (0.6% obs/m) by number of holes in overall system pipe network then add the recorded average ambient peak level to derive at the Alert threshold.

2. Air sampling detection apparatuses which permit scheduling of sensitivity thresholds by date and/or time shall be programmed to use the same sensitivity thresholds specified herein 24 hours a day, 7 days a week.
3. Ten days prior to the test, a data logger is to be installed to monitor background ambient conditions for each zone. Document tests.
4. Time delays are intended to mitigate nuisance signals as a result of transient conditions. The following time delays shall be programmed into each detector apparatus:

   i. Fire 2: 30 seconds.
   ii. Fire 1: 15 seconds.
   iii. Action: 30 seconds.
   iv. Alert: 30 seconds.
   v. Airflow fault delay: 60 seconds.
   vi. Scan Delay (four zone air sampling detector apparatuses): 15 seconds.

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11.8 Final Acceptance Testing and Commissioning

1. The following list of documents shall be assembled before commissioning test begins:
   i. The HSSD system design criteria, including Sequence of Operation.
   ii. Programed sensitivities, alarm thresholds, and time delays.
   iii. As-Built drawings showing the sampling pipe network
   iv. The HSSD computer-based design calculation
   v. Required local authority test report forms.
   vi. All materials and equipment necessary to complete the commissioning test such as canned test smoke, stop watch, ladder, test forms, computer with design software for setting verification).
2. The following items need to be verified to complete the test in accordance with the applicable code and manufacturer's recommendations:
   i. Check the electrical and signal cabling. Refer to the manufacturer's installation, wiring, and cabling requirements for detailed and specific information. Verify the completed system is free from grounded or open circuits.
   ii. Verify that the HSSD detector(s) and the associated power supply(s) are monitored by graphic workstation.
   iii. Visually inspect the sampling pipe network to ensure the installation complies with the as-built drawings and to verify the pipe is supported properly.
   iv. Ensure the protected room or area is in its normal operational condition in terms of airflow, temperature, and cleanliness.
   v. Prior to the test, and in accordance with NFPA 72 or equivalent, verify the system is placed in testing mode and all occupants are notified.
   vi. Typically, minimum three teams shall perform the commissioning test of the HSSD system, with one introducing canned smoke into the sampling hole, one at the detector location to acknowledge when smoke is detected, and the other one at monitoring systems. The result of the travel times are documented on the test form.
   vii. Perform airflow, fault, and sensitivity tests on the complete installed system. Perform time-delay tests for detector response using a data logger. Adjust the time-delay settings to avoid unwanted alarms. Document tests.
   viii. Perform tests on system response to verify it is within the specified range. Expose canned smoke, approved by Owner for use in the area, in the air stream a minimum of 6 feet upstream from the farthest sampling point and record system response time and the duration of the canned smoke release to trigger low and high alarms. Check for sampling pipe integrity and pipe blockage. Correct any problems and repeat test.
   ix. During normal operating conditions, should the detector assembly's sensitivity be exceeded by the contaminants in the air, it is to be replaced with a detector that is less sensitive.

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x. Defects to be corrected at once and the test reconducted at no additional cost to Micron.

3. The HSSD system commissioning should include testing and verifying that all programmable relay outputs function as designed. Interlock test shall be carried out to meet the Sequence of Operation.
4. All test results must be recorded in accordance with local code and regulations, as well as within manufacturer's recommendations. At the minimum, the commissioning form should contain the following information:
   i. Micron information: Building name and Address.
   ii. Date of the test.
   iii. Installing contractor's information, including name, address, and contact telephone number.
   iv. Identification of the rooms being protected with the HSSD system.
   v. Witnessing/approving authority information.
   vi. The HSSD information:
      a. Detector serial number.
      b. Sensitivity
      c. % obs/m or % obs/ft
      d. Thresholds, with day, night, and weekend settings.
      e. Time delays.
      f. Type of detection system.
      g. Number of sampling pipes and sampling holes.
      h. Hole size.
      i. Testing result related to transport time.
      j. Signatures of all the persons attending the test (agents performing the test, contractor, Micron representatives, and Authority representative)
5. The commissioning teams with Micron and Authority representatives should be completely satisfied with all the results from the commissioning tests. They should also agree that all testing results meet the local codes and regulations and manufacturer's recommendations for the type of system being tested, and that testing results confirm the calculated hole and transport times calculated with the design software. The final acceptance of the tested HSSD system should be a complete and signed copy of commissioning test form, a copy of the as-built drawings, and Operation Maintenance Manual.

11.9 TRAINING

Provide the Owner a minimum of 4 hours of instruction for six technicians in the operation and maintenance of the system and test equipment.

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12.0 Document Control

12.1 Approval: GLOBAL_FAC_GFTT_APPR

12.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10W_FAC_NOTIFY
FCG_F10N_FAC_NOTIFY
FCG_MTTW_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
MTB_FAC_ALL_NOTIFY

12.3 Final Viewer Access

All Micron Team Members at all Sites

12.4 Retention

Adherence to the requirements specified in the **Global Facilities – Document Control Procedures** and **Global Quality Records Retention Standard**.

12.5 Review

Annually

13.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	10/30/2020

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Appendix A – Deviation List

Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall be the following:

Site: ______________________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT

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